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# VOID FRACTION IN A SINGLE CHANNEL SIMULATING ONE SUBCHANNEL OF A PWR FUEL ASSEMBLY

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**ABSTRACT** - The void fraction measurement tests for pressurized water reactor (PWR) fuel assemblies have been performed since 1987 under the sponsorship of the Ministry of International Trade and Industries (MITI) as one of the Japanese national projects. These tests include the single channel tests performed as a preliminary run for the rod bundle tests under various steady state and transient conditions.

Gamma-ray attenuation method is used to measure subchannel void fractions under the PWR thermal-hydraulic conditions. The single channel experiment with the test section simulating an inner subchannel was performed. And, the obtained void data have been processed and compared with predictions by the void calculation methods consisting of wall voidage model, detached void model, subcooled boiling model and void-quality correlation. This paper describes the experimental information, the results of void fraction measured in the single channel steady state test and the comparison of experimental results with the predictions.

## 1. INTRODUCTION

The void fraction in a PWR core is negligible under the normal operation conditions, but, becomes significant in high powered fuel assemblies under anticipated transients or accident conditions. The void generation in the fuel assemblies provides the redistribution of the coolant flow throughout the core, and reduces the fission power due to the void reactivity feed back mechanism. As such, the void behavior affects not only the thermal-hydraulic characteristics but also the nuclear characteristics. Therefore, the void behavior in the fuel assemblies is one of the most important factors from the viewpoint of reactor safety even in a PWR nuclear power plant.

The experimental studies on the void fraction in single channels were performed by gamma-ray techniques [1-4], and those in rod bundles were performed by gamma-ray techniques [5,6] or by neutron techniques [7,8]. However, there are no subchannel averaged void data under steady state and transient conditions corresponding to PWR upset conditions. The subchannel averaged void data is essential for the PWR thermal-hydraulic analysis. Therefore, the void fraction measurements both in simulated multi-rod bundles and in a single channel simulating one subchannel of a PWR fuel assembly were planned and have been performed [9,10]. The purpose of these experiments is to obtain the data to verify the current PWR void prediction method or to develop a more suitable prediction method.

The single channel tests are preliminary for the rod bundle tests under steady state and transient conditions simulating anticipated transients or accident conditions in a PWR. Test sections simulating one of the subchannels in a PWR fuel assembly were used in these single channel tests. The experiments with the test section simulating one of the inner subchannels of PWR fuel assembly were performed under steady state conditions and transient conditions [11].

The density of two-phase flow was measured by gamma-ray attenuation method and converted to void fraction. The channel averaged density can be obtained from the density distribution measured by the gamma-ray computed tomography (CT) scanner system under steady state conditions. The required measuring time of the CT scanner system is about two hours. Therefore, the stationary chordal measurements were also performed to get the adequate data points. The chordal averaged values are corrected to the channel averaged values by the relationship between the channel averaged value and the chordal averaged value. This relationship was obtained by the CT measurements and the chordal measurement under the same experimental conditions. The obtained correlation will also be applied to the evaluation of bundle test data measured by the chordal measuring system.

This paper presents the channel averaged void data obtained from the steady state experiments of inner subchannel test section performed systematically under the wide parameter range, and the assessment of the existing void prediction methods which may consist of wall voidage model, detached voidage model, subcooled boiling model and void-quality correlation. The obtained data are mainly used to assess the existing void fraction prediction methods. It is needless to say that such effect as flow redistribution between neighboring subchannels should be taken into consideration for the data evaluation of the rod bundle experiments.

## 2. TEST FACILITY AND CONDITIONS

### 2.1 Test Loop

This experiment was performed in a test loop designed to cover the conditions during anticipated transients or postulated accidents in a PWR core including such transients as flow reduction, depressurization, and inlet temperature increase. The test loop is used for both single channel tests and rod bundle tests under steady state and transient conditions.

The system diagram of this loop is shown in Figure 1. The loop consists mainly of two systems, a circulation system and a cooling system. The circulation system supplies high temperature pressurized water at fixed temperature, pressure and flow rate for each test section. The cooling system removes the heat added to the water in the loop and controls temperature and pressure for the given conditions.

This loop has two test sections, a single channel test section and a rod bundle test section. In the case of single channel tests, the rod bundle test section is used as a bypass line of the flow fed by the circulation pump with a large flow capacity relative to the single channel test section.

The flow rate is controlled by an oil pressure driven control valve and measured by a Venturi tube flow meter in the flow control section upstream of each test section. The water temperature is controlled by a preheater and measured by a calibrated platinum resistive thermometer (RTD) at the inlet of the test section. The pressure is controlled by adjustment of electric heater power and subcooled water spray flow rate of the steam drum. The test section inlet pressure is measured by a high precision quartz manometer.

Degassed distilled water is supplied to the loop as the coolant and the main equipment is made of stainless steel to prevent water fouling. The design pressure is 19.2 MPa and the design temperature is 362 ° C. The steady-state test parameter ranges cover:

| | | |
|---|---|---|
| Pressure : | 4.9 - 16.6 | MPa |
| Inlet Temperature : | 140 - 345 | ° C |
| Mass Velocity : | 550 - 4150 | $kg/m^2 s$ |

The transient parameter ranges are as follows.

| | | |
|---|---|---|
| Depressurization : | - 0.03 | MPa/s |
| Temperature Increase : | 1 | ° C/s |
| Flow Reduction : | - 25 | %/s |

Accuracies of the test parameters are estimated as:

| | | |
|---|---|---|
| Pressure : | ± 1 | % |
| Inlet Temperature : | ± 1 | ° C |
| Flow Rate : | ± 1.5 | % |
| Test Section Power : | ± 1 | % |

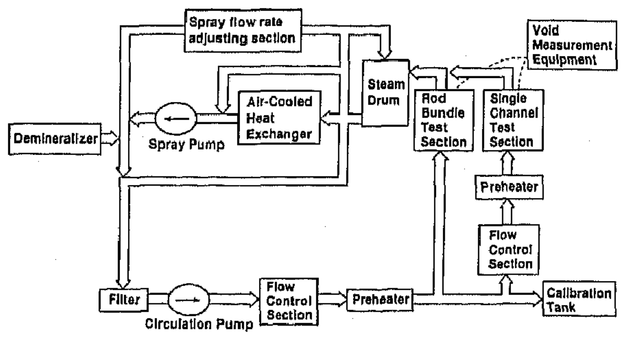


Figure 1 System Diagram of Test Loop

### 2.2 Test Section

Figure 2 shows the single channel test section. The single channel test section consists of a pressure vessel made of titanium alloy, a test channel and electrodes supplying direct current electric power for the heater element. The void measuring system is set at the measuring section. The wall thickness of the pressure vessel where the gamma-ray beam transmits is cut down to reduce the gamma-ray attenuation in the wall.

The test channel simulates an inner subchannel of a PWR fuel assembly. This channel consists of four skin heater elements of Inconel alloy 600 simulating the quarter sections of four actual heater rods, and a ceramic insulator to form a flow channel and insulate the heater elements electrically from the other structures.

The outer diameter of the simulated rod is 9.5 mm, the rod pitch is 12.6 mm, and the rod gap is 3.1 mm. The other cross-sectional dimensions of the test channel are shown in Figure 2. The total flow area is 107.1 $mm^2$ . The effective heated length is 1.5 m where the void measuring section is set near the top end. Each heater element has the same power output and the axial power distribution is uniform.

### 2.3 Void Measuring System

The density of two-phase flow is measured by gamma-ray attenuation method and converted to the void fraction. The mixture density is deduced from the count data given by the gamma-ray measuring system. The count rate for the attenuated beam N is related to the mixture density $\rho$ by the next equation.

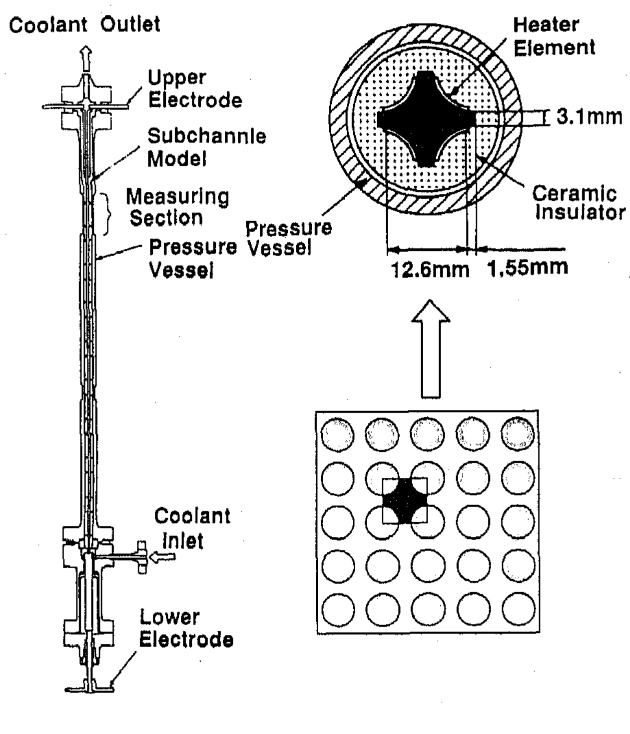


Figure 2 Single Channel Test Section

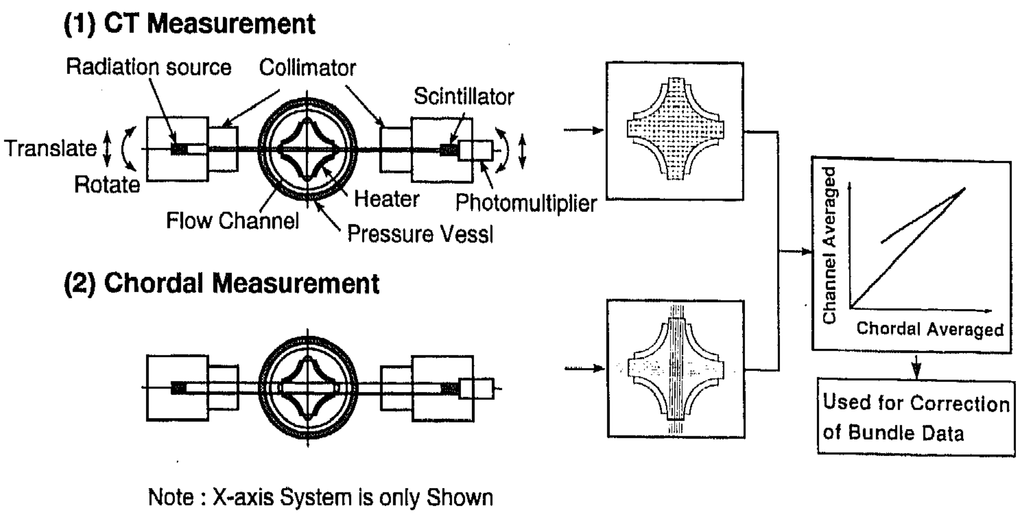


Figure 3 Void Measuring Method

$$N = N_0 \, \mathrm{EXP} \, (-\rho \mu L) \quad (1)$$

where $N_0$ : count rate for incident beam,
$\mu$ : mass attenuation coefficient
and $L$ : attenuation length.

The void fraction $\alpha$ is related with the mixture density $\rho$ of the two-phase flow as follows.

$$\rho = \alpha \rho_g + (1-\alpha) \rho_L \quad (2)$$

where $\rho_g$ : density of gas ,
and $\rho_L$ : density of liquid.

Rearrangement leads to

$$\alpha = \frac{\rho_L}{\rho_L - \rho_g} - \frac{1}{\rho_L - \rho_g} \rho \quad (3)$$

The actual channel averaged density can be obtained from the measurements with a gamma-ray CT scanner system under the steady state conditions as shown in Figure 3(1). However, this gamma-ray CT scanner system requires the long measuring time, about 2 hours. Therefore, the stationary wide beam densitometer systems are also used in this test to get the adequate data points due to its shorter measuring time, 100 seconds.

The wide gamma-ray beam densitometer system gives x-direction and y-direction chordal averaged measurements as shown in Figure 3(2). This system can give the time-averaged densities during each 0.1 seconds continuously even in transient conditions. However, this system does not enable us to measure over the whole region of the channel.

A correlation between the chordal averaged density and the actual channel averaged density is required to obtain the channel averaged value from the measured results by the wide gamma-ray beam densitometer system. So, the required correlation must be obtained from a comparison between the measurements of the subchannel averaged density by the CT scanner system and the chordal averaged density by the wide gamma-ray beam densitometer system.

The void measuring system for single channel tests is shown in Figure 4. This system consists of gamma-ray sources, detection parts, collimators, translate/rotate tables and signal processing units. The specifications of this system were optimized to reduce the measurement error.

Radio isotope cesium-137 having the half life of about 30 years sealed in a capsule is used as a stable gamma-ray source and mounted in a container made of tungsten. The activity of each gamma-ray source is $3.7 \times 10^{11}$ Bq. The collimators are placed in front of the sources and the detectors to collimate the gamma-ray beam and eliminate scattered rays. The detection part consists of a scintillator, a light guide, a photomultiplier and a preamplifier. A crystal of cerium-doped gadolinium silicate (GSO) is used as a scintillator because its scintillation decay time is short enough to measure gamma-ray with a large efficiency at high count rates, up to 300,000 cps, which was required to reduce the error due to photon statistics. The detection part is put in a magnetically shielded temperature controlled box to minimize the output drift.

The output signal of the preamplifier is transferred to the signal processing unit working in the pulse counting mode. The input signal drift of the signal processing unit is canceled by a spectrum stabilizer feedback circuit so that the voltage detector pulse corresponding to the photoelectric absorption of a 661 keV photon should be constant at the counter. Pulse pileup phenomena observed due to dead time of the count meter in the pulse counting mode is compensated based on the paralysable model [12] in the count data processing. In addition, measurements are

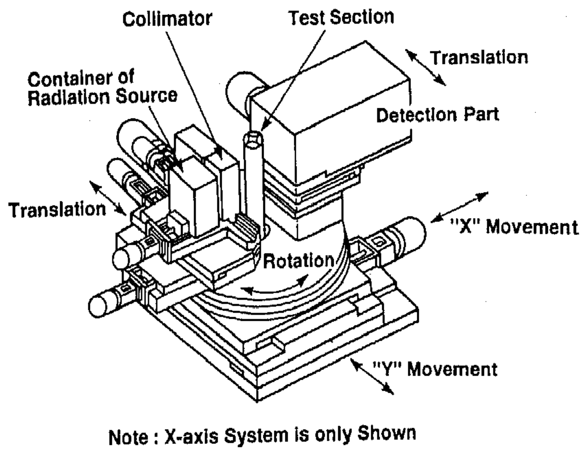


Figure 4 Void Measuring System

effectively calibrated by data obtained for an empty condition of the test section and a water filled condition at room temperature and for non-boiling conditions at high temperatures.

When the measuring system is applied as a chordal densitometer system, the gamma-ray sources and detection parts are set at stationary location by adjusting the translation tables so that each gamma-ray beam of x-direction system and y-direction system transmits through the center of each rod gap. In this case, the width of collimator slit is determined to be 2.5 mm to avoid the interaction of gamma-ray beam with heating elements. The measuring time is set to be 100 seconds for the steady state tests.

The measuring system also works as a CT scanner system by translating and rotating the gamma-ray sources and the detection parts. Collimators having 1.0 mm width slits are used in this case. Two measuring systems, an x-direction system and a y-direction system, are used to reduce the required measuring time. The system has 33 rotation locations. The number of translation locations is 33 at each rotation location. The translation distance is set at 0.5 mm and the diameter of the region of interest is 16 mm. Void/density distribution and channel averaged value are obtained from the distribution of the linear attenuation coefficients that depend on image reconstruction using a filtered back-projection algorithm [13]. The minimum spatial resolution of this CT scanner is 0.5 mm. A sufficient measuring time at each measuring location is chosen to avoid the effect of motion of void and to obtain the time averaged values. The required time is determined to be 5 seconds after confirming that the measured channel averaged values obtained from several different measurement times are almost constant and do not depend on the measuring time under various flow conditions. Therefore, it takes about 7200 seconds including the required time for translate/rotate movement to complete the measurement at one test condition.

The accuracy of the void fraction measurement was assessed by taking account of the photon statistics of gamma-ray, the errors from the compensation method for pulse pileup phenomena, the errors of spectrum stabilizer for canceling the drift of the detectors or the signal processing unit, and density measuring errors at the calibration conditions due to process parameter measuring errors. The obtained total errors of count rate are as follows:

(1) Chordal Measurement : less than $\pm$ 0.3 %

(2) CT Measurement : less than $\pm$ 0.3 %

The density/void fraction of chordal measurement is obtained by an algebraic calculation using Equations (1) and (3) from the count data [14,15]. Equation (3) gives

$$\frac{\Delta \alpha}{\Delta \rho} = -\frac{1}{\rho_L - \rho_g} \tag{4}$$

Rearrangement leads to

$$\frac{\Delta \alpha}{\alpha} = \left\{ 1 - \frac{\rho_L}{\alpha\,(\rho_L - \rho_g)} \right\} \frac{\Delta \rho}{\rho} \tag{5}$$

Rearrangement of equation (1) leads to

$$\Delta \rho = -\frac{1}{\mu L}\,\frac{\Delta N}{N} \tag{6}$$

Equations (4), (5) and (6) can be applied to the wide-beam chordal measurement directly. The evaluated absolute measuring error of chordal averaged void fraction is less than $\pm$ 4 % for the steady state measurements.

On the other hand, the filtered-back projection algorithm is applied as the reconstruction algorithm of CT scanner system to obtain the void fraction distribution. Therefore, the accuracy of channel averaged void fraction by the CT scanner was examined by numerical simulation methods, applying the count rate error obtained in the above assessments. It was estimated that the absolute measuring error of channel averaged void fraction by the CT scanner is less than $\pm$ 3%. In the averaging process, the fluid density in the picture element on the solid-fluid interface is deduced from the measured element density, the area ratio occupied by the solid wall and the solid density. The measuring error of chanel averaged value is also confirmed to be within above value by measuring the mock-up simulating a structure of test section and an acrylic resin phantom as a model of two-phase flow with such data process[9].

In all tests, measured fluid densities in single phase conditions were compared directly with the theoretical values obtained from a steam table, and it was confirmed that they agree within $\pm$ 1.5%.

## 3. TEST RESULTS AND DISCUSSIONS

### 3.1 Preliminary Results

The single channel test using the test section simulating an inner subchannel was performed under the test conditions as shown in Table 1. Both the chordal averaged measurement and the channel averaged measurement were preformed under the test conditions with an asterisk in this table. The chordal averaged density/void-fraction was measured by the wide gamma-ray beam densitometer systems and the channel averaged density/void-fraction was measured by the CT system.

Figure 5 shows the relation between x-direction chordal density and y-direction chordal density, both of which are obtained simultaneously by the wide gamma-ray beam densitometer systems as shown in Figures 3 and 4. It is obvious that both densities agree well and the measured density of the test section simulating the inner subchannel of a PWR does not depend on the measuring direction. So, the averaged values of these two densities were used as the typical value of channel, the chordal averaged density.

The correlation between the chordal averaged density and the channel averaged density was introduced to estimate the channel averaged density from the chordal averaged density. Figure 6 shows the comparison of the density between the wide beam densitometer and the CT scanner. Solid circles represent the data obtained under high pressure conditions and hollow circles represent the data obtained under relatively low pressure conditions.

The wide beam densitometer gives smaller density which means larger void fraction than the CT scanner at the bulk boiling region under relatively low pressure conditions. On the other hand, the discrepancy of the two values is not significant under the high pressure conditions. These results may mean that the system pressure makes the void fraction distribution to be different. Figure 7 shows the void fraction distribution measured by the CT scanner system. The left reconstructed image represents the relatively high pressure 14.7 MPa case and the right reconstructed image represents the relatively low pressure 9.8 MPa case. Each case has the same mass velocity 5x10 $^{6}$ kg/m $^{2}$ h and the same heat flux 122 W/cm $^{2}$ . The channel averaged void fraction of each case is almost same, 50%. However, the different void fraction distributions are observed. For the relatively high pressure condition, many voids remain near the heated wall. On the other hand, the voids concentrate in the central part of the channel for the relatively low pressure condition.

The correlations between the chordal averaged value and the channel averaged value given by the best fit curves as shown in Figure 6 were introduced respectively for the high pressure conditions and the low pressure conditions. The uncertainty of this correlation is evaluated to be less than 18 kg/m $^{3}$ for steady state conditions.

### 3.2 Single Channel Steady State Test Results

The channel averaged void fraction data were obtained from the CT measurement and the chordal measurement corrected by the correlation between the channel averaged value and the chordal averaged value. The test results were

Table 1 Test Matrix

| Pressure (MPa) | Mass Velocity (x10⁶ kg/m²h) | Power (kW) | Inlet Temperature (℃) |
|---|---|---|---|
| 16.6 | 15 | 60 | 335 |
| | 11 | 80 | 315 |
| | 11 | 50 | 330*, 335*, 340* |
| | 5 | 40 | 335 |
| 14.7 | 15 | 100 | 305 |
| | 15 | 80 | 310*, 320* |
| | 11 | 90 | 295*, 300* |
| | 11 | 70 | 300*, 310 , 320* |
| | 11 | 60 | 300*, 305 , 310* 315*, 320*, 325*, 330* |
| | 8 | 50 | 315 |
| | 5 | 80 | 250 |
| | 5 | 60 | 270*, 285*, 300* |
| | 5 | 20 | 320 |
| 12.3 | 15 | 70 | 315 |
| | 11 | 90 | 280 |
| | 11 | 60 | 295*, 310*, 320* |
| | 5 | 40 | 295 |
| 9.8 | 11 | 100 | 260 |
| | 11 | 70 | 275*, 305* |
| | 8 | 60 | 300 |
| | 5 | 80 | 215*, 250* |
| | 5 | 60 | 210*, 220 , 230*, 240*, 255*, 270*, 290* |
| | 2 | 20 | 255*, 285* |
| 7.4 | 11 | 80 | 260 |
| | 5 | 90 | 165 |
| | 5 | 50 | 220*, 245*, 265* |
| | 2 | 20 | 245 |
| 4.9 | 8 | 70 | 215 |
| | 5 | 80 | 165*, 200* |
| | 5 | 50 | 190*, 205*, 240* |
| | 2 | 20 | 205*, 240* |

* : CT Meurement anChordal Measurement

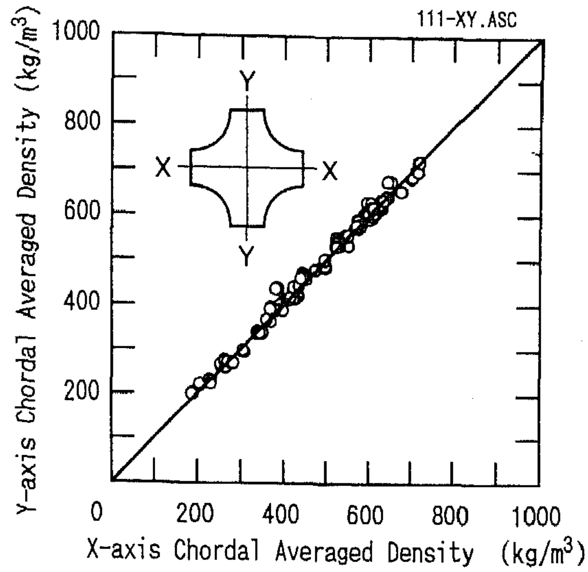


Figure 5 Relation between X-direction and Y-direction Chordal Density [11]

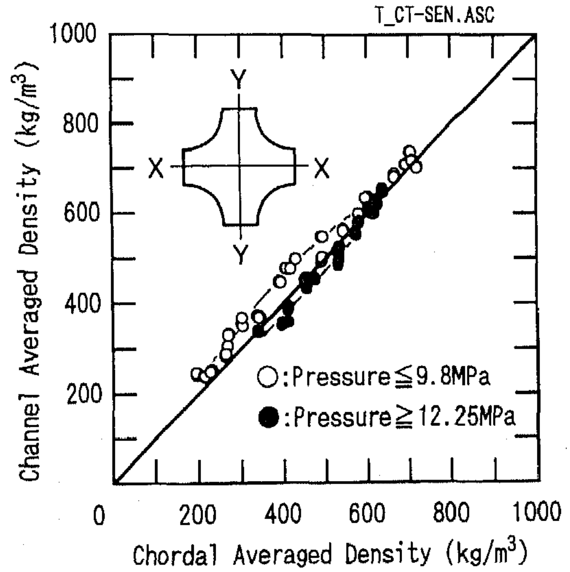


Figure 6 Relation between Chordal Averaged Density and Channel Averaged Density [11]

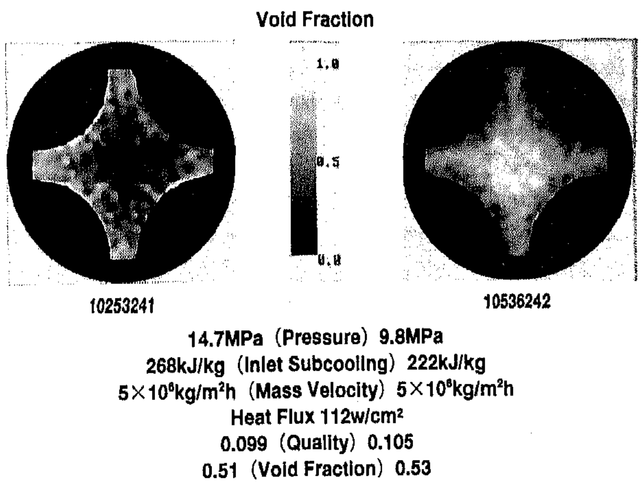


Figure 7 Void Fraction Distribution by CT Scanner

compared with various void prediction methods as shown in Table 2 [16-18]. A void prediction method may consist of a wall voidage model, a detached void model, a subcooled boiling model and void-quality correlation. The models in parentheses of Table 2 were assumed as optional variations for the original prediction methods.

In subcooled boiling conditions, a liquid temperature is required to calculate the void fraction from the measured mixture density. Unfortunately the liquid temperature cannot be determined from the measurements. So, it is necessary to introduce an adequate assumption. In the present evaluation, the same void prediction models are used to determine the liquid temperature required for calculating the void fraction from the measured density values. However, the void fraction calculated from the measured mixture density depends only slightly on the models used for the determination of liquid temperature. The vapor is assumed to be at saturated temperature.

Figures 8 and 9 represent the relations between the void fraction and the thermal equilibrium quality. Figure 8 is the relatively lower pressure, 9.8 MPa, case. Figure 9 is the relatively high pressure, 14.7 MPa, case. The hollow symbols represent the measured void fraction data, and the solid lines represent the predictions by the prediction method shown in Table 2. The comparison between the measured data and some predictions shows that the prediction method Lahey(1) gives the better predictions for the relatively lower pressure case and the prediction method Lahey(3) gives the better predictions for the relatively high pressure case. The prediction method Lahey(1) consists of Levy model as the void detach model, Lahey model as the subcooled boiling model and Zuber-Findlay drift flux model as the void quality correlation. On the other hand, the prediction method Lahey(3) uses the homogeneous model as the void-quality correlation instead of the drift flux model.

Figure 10 contains the data measured under the relatively lower pressure conditions, 4.9 MPa, 7.4 MPa and 9.8 MPa. The void fraction ranges up to 70 %. These data are compared with the predictions. This figure also shows that the prediction method Lahey(1) gives the better predictions for the relatively lower pressure case than the other methods assessed in this paper. However, every method overpredicts the void fraction under the lower flow conditions. Figure 11 shows the comparison between the data measured under the relatively high pressure conditions, 12.3 MPa, 14.7 MPa and 16.6 MPa, and the predictions. The void fraction ranges from a few % to 60 %. It is also shown that the better predictions are obtained by the prediction method Lahey(3).

Figures 12 and 13 represent the measured data under the relatively lower pressure conditions with the predictions by Lahey(1) as the relation between the void fraction and the thermal equilibrium quality. In Figures 14 and 15, the measured void fraction data obtained under the relatively high pressure conditions are plotted against the thermal equilibrium quality with the predictions by Lahey(3).

It is obvious from Figure 7 that the void distribution under the high pressure condition is different from under the low pressure condition. The bubbles concentrate in the central part of the channel under the low pressure condition. On the other hand, they remain near the heated wall under

Table 2 Applied Void Prediction Method

| Total Void Prediction Method | Wall Voidage Model | Void Detach Model | Subcooled Boiling Model | Void-Quality Correlation |
|---|---|---|---|---|
| Bowring[16] | Maurer | Bowring | Bowring | Smith |
| Ahamad[17] | – | Ahmad | Ahmad | Ahmad |
| Lahey[18](1) | – | Levy | Lahey | Zuber-Findlay |
| Lahey[18](2) | (Maurer) | Levy | Lahey | Zuber-Findlay |
| Lahey[18](3) | (Maurer) | Levy | Lahey | (Homogeneous) |

( ):model used in this calculation

the high pressure condition even void fraction is as high as 50 %.

The bubbles flow in the low velocity region near the wall under the high pressure condition. Therefore, it is estimated that the homogeneous flow model agrees well under this condition.

Under the low pressure condition, the bubbles of bubbly or plug flow regime flow in the high velocity central region of the channel. So, the drift-flux model is estimated to be applicable under the low pressure condition.

## 4. CONCLUSION

The single channel test with a test section simulating an inner subchannel of a PWR was performed as one of the tests planned in the void fraction measurement experiments for PWR fuel assemblies. The major conclusions in this paper are as follows.

(1) An overview of all the comparisons between the measured void fraction and the predictions by some prediction methods gives the next results. The void fraction under the relatively lower pressure conditions below 9.8MPa is well predicted by Lahey(1) prediction method which uses the drift-flux model as the void-quality correlation. On the other hand, the modified prediction method Lahey(3) taking the homogeneous flow assumption as a void-quality correlation instead of the drift-flux model gives the better predictions under the relatively high pressure conditions above 12.3MPa.

(2) The obtained correlation between the channel averaged measurement and the chordal averaged measurement will be applied to evaluation of the bundle test data measured by the chordal measuring system.

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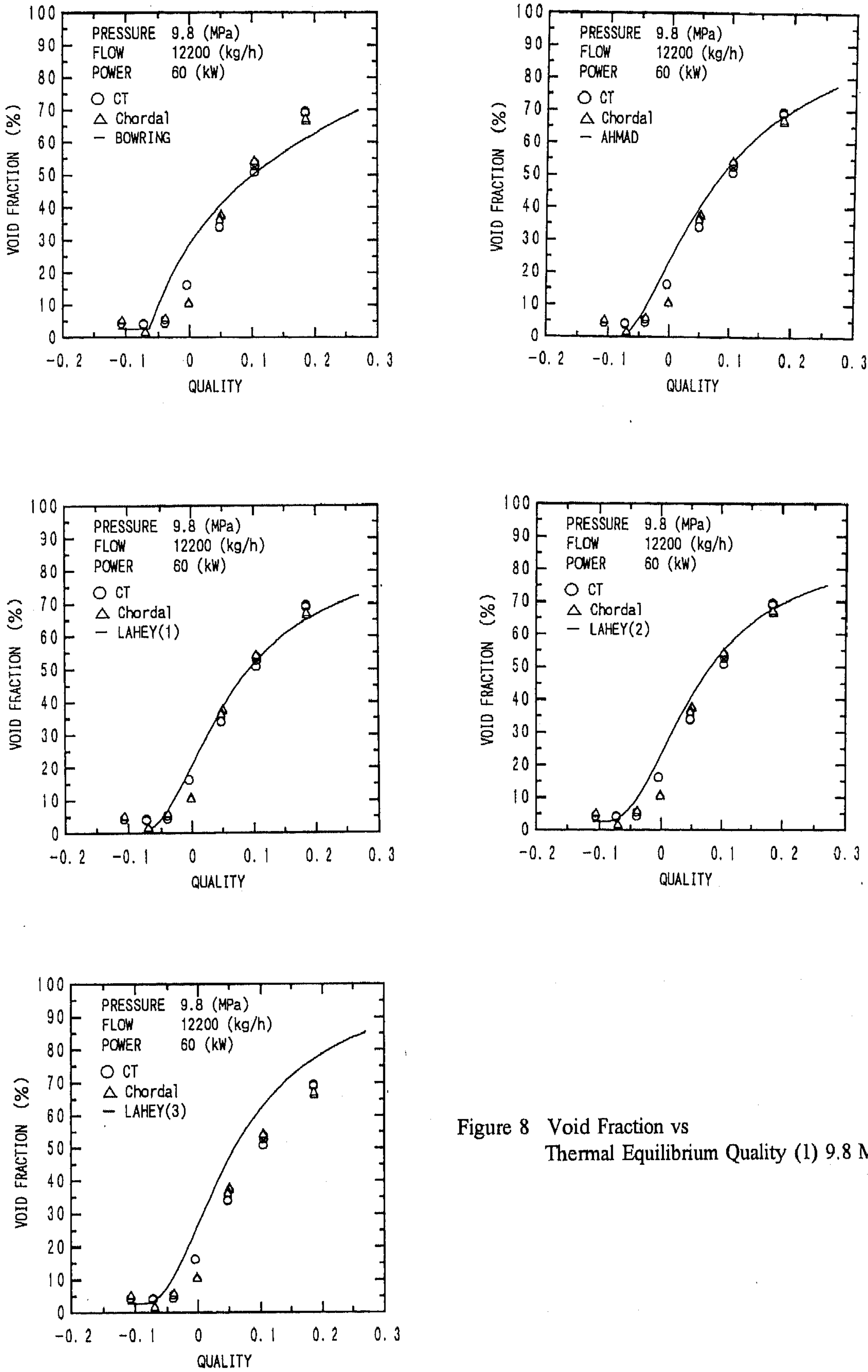



Figure 8 Void Fraction vs Thermal Equilibrium Quality (1) 9.8 MPa

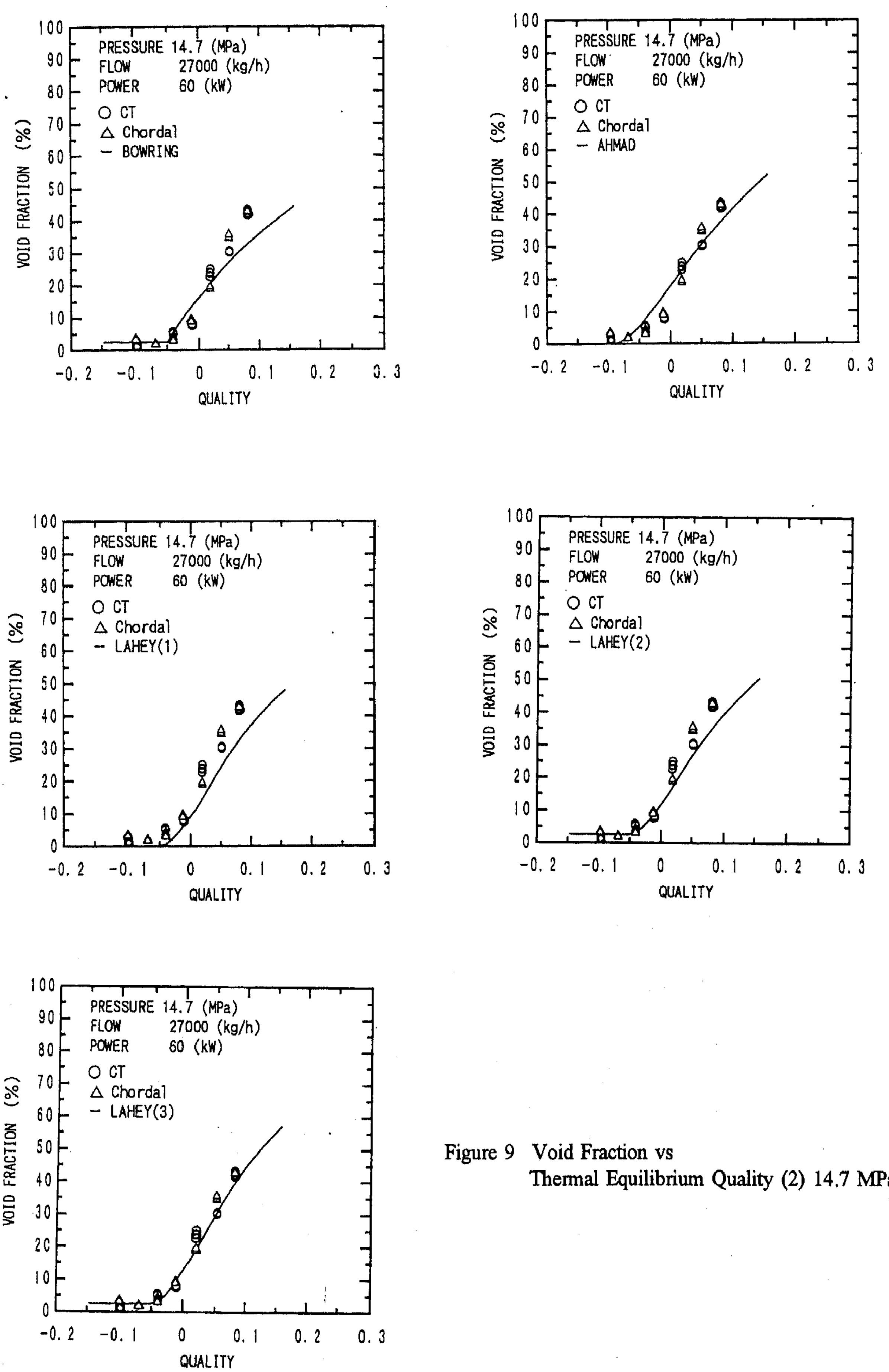
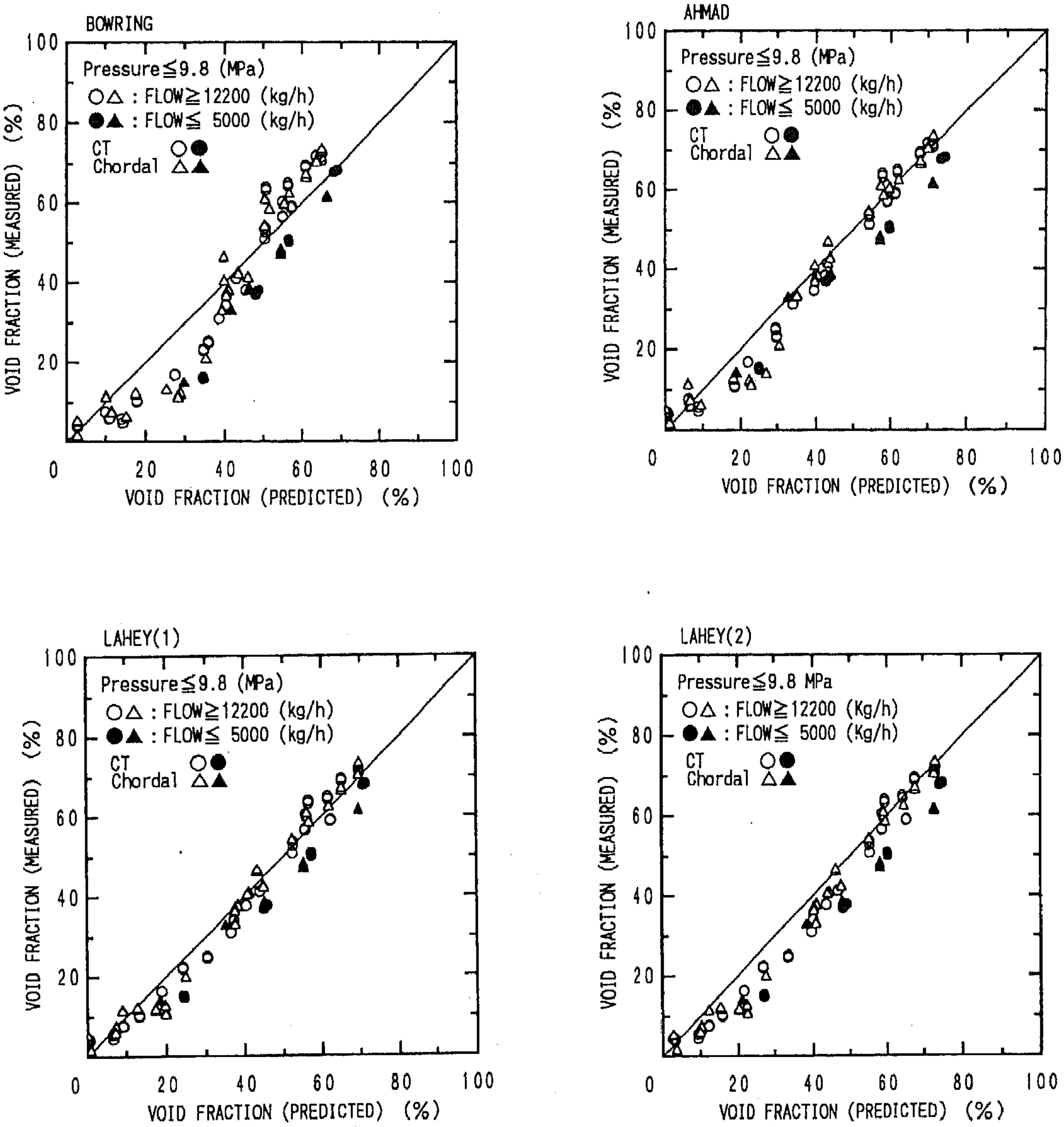



Figure 9 Void Fraction vs
Thermal Equilibrium Quality (2) 14.7 MPa



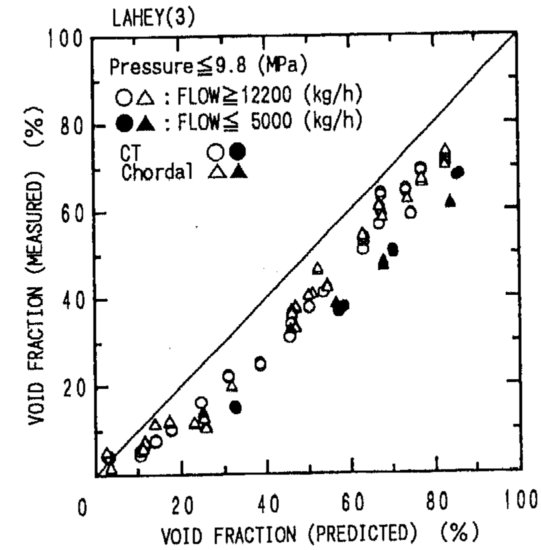




Figure 10 Comparison of Measured Void Fraction with Predictions (1) Low Pressure Cases

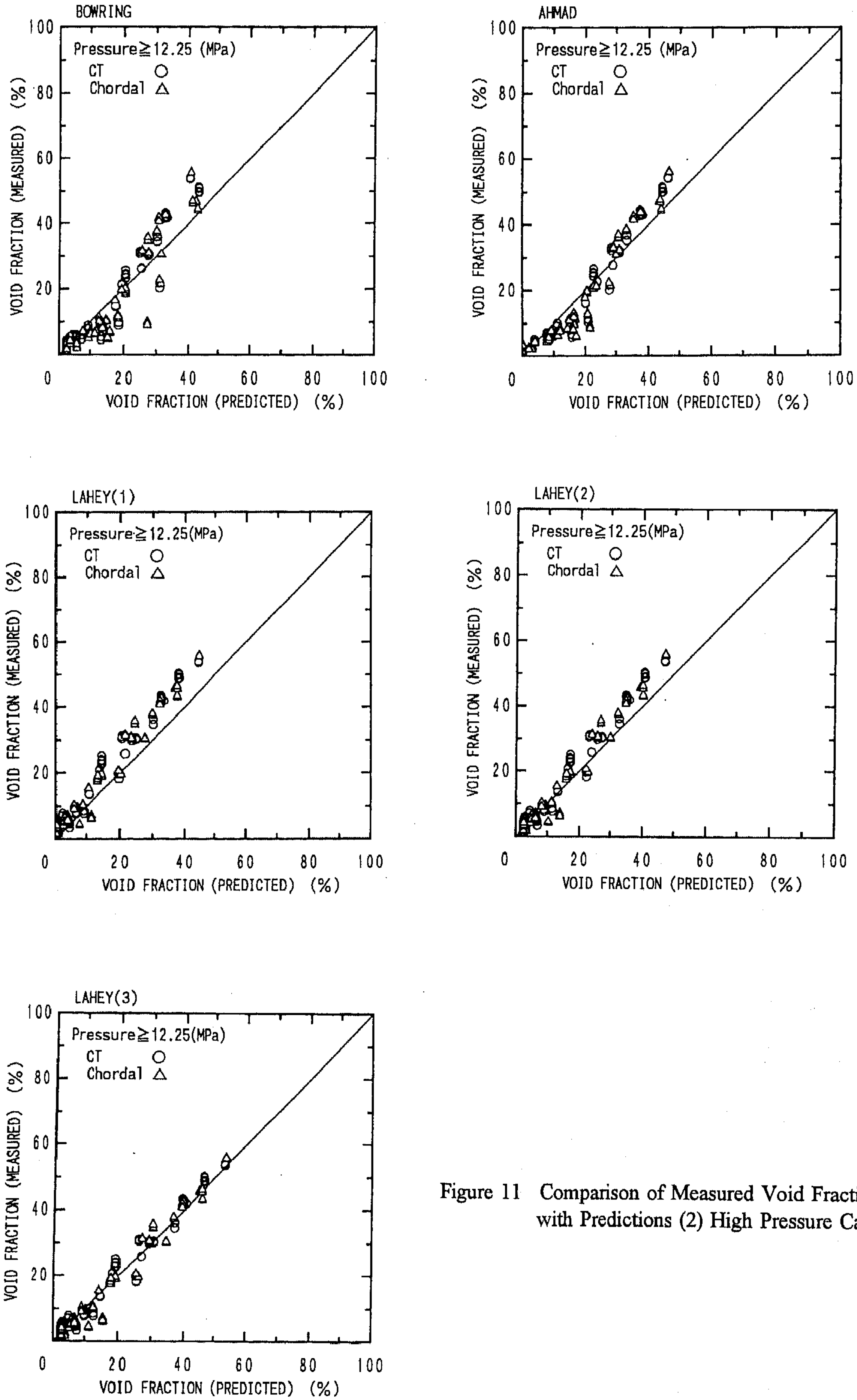



Figure 11 Comparison of Measured Void Fraction with Predictions (2) High Pressure Cases

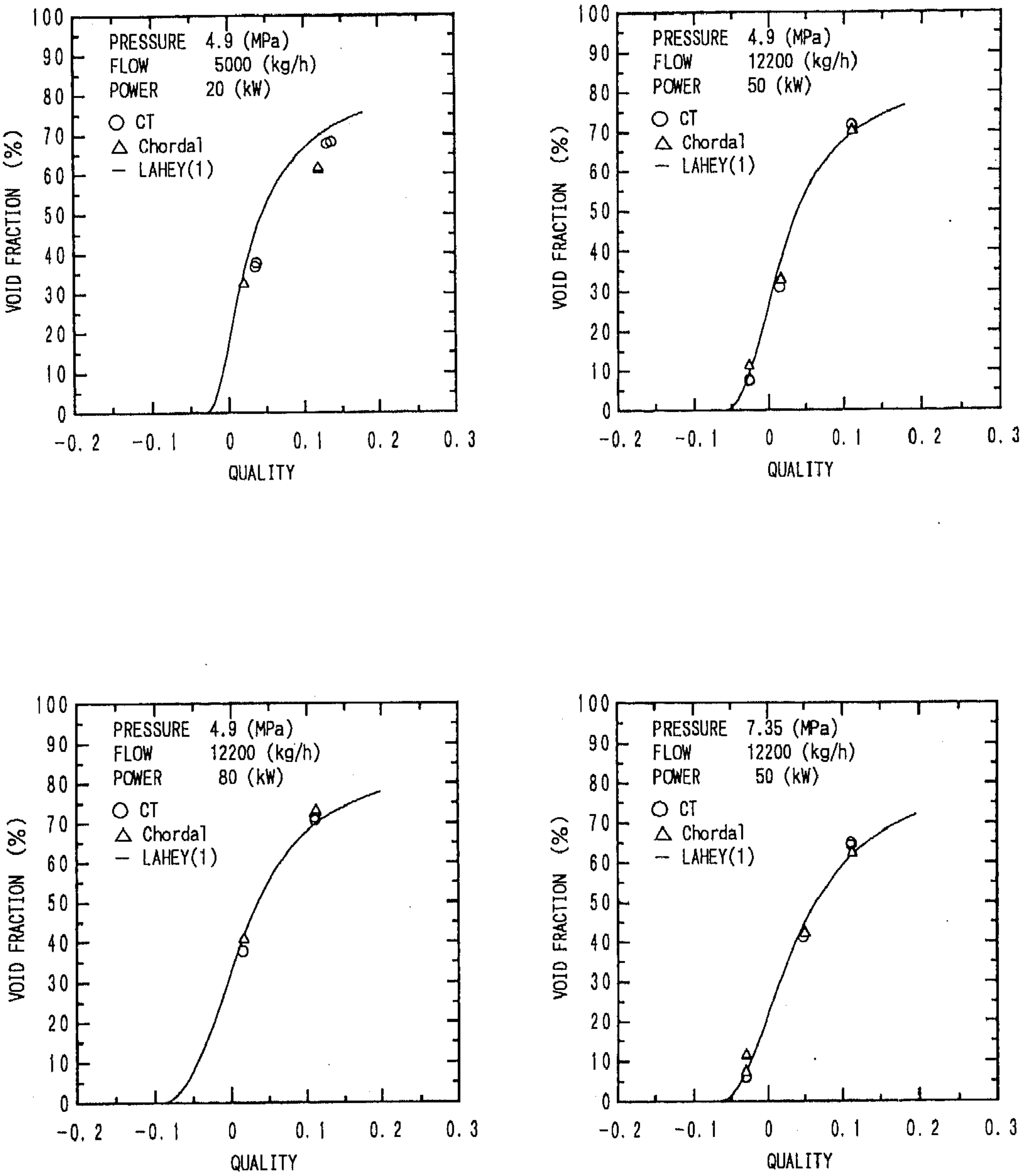



Figure 12 Measured Void Fraction vs Thermal Equibrium Quality with Best Prediction (1) 4.9-7.35 MPa

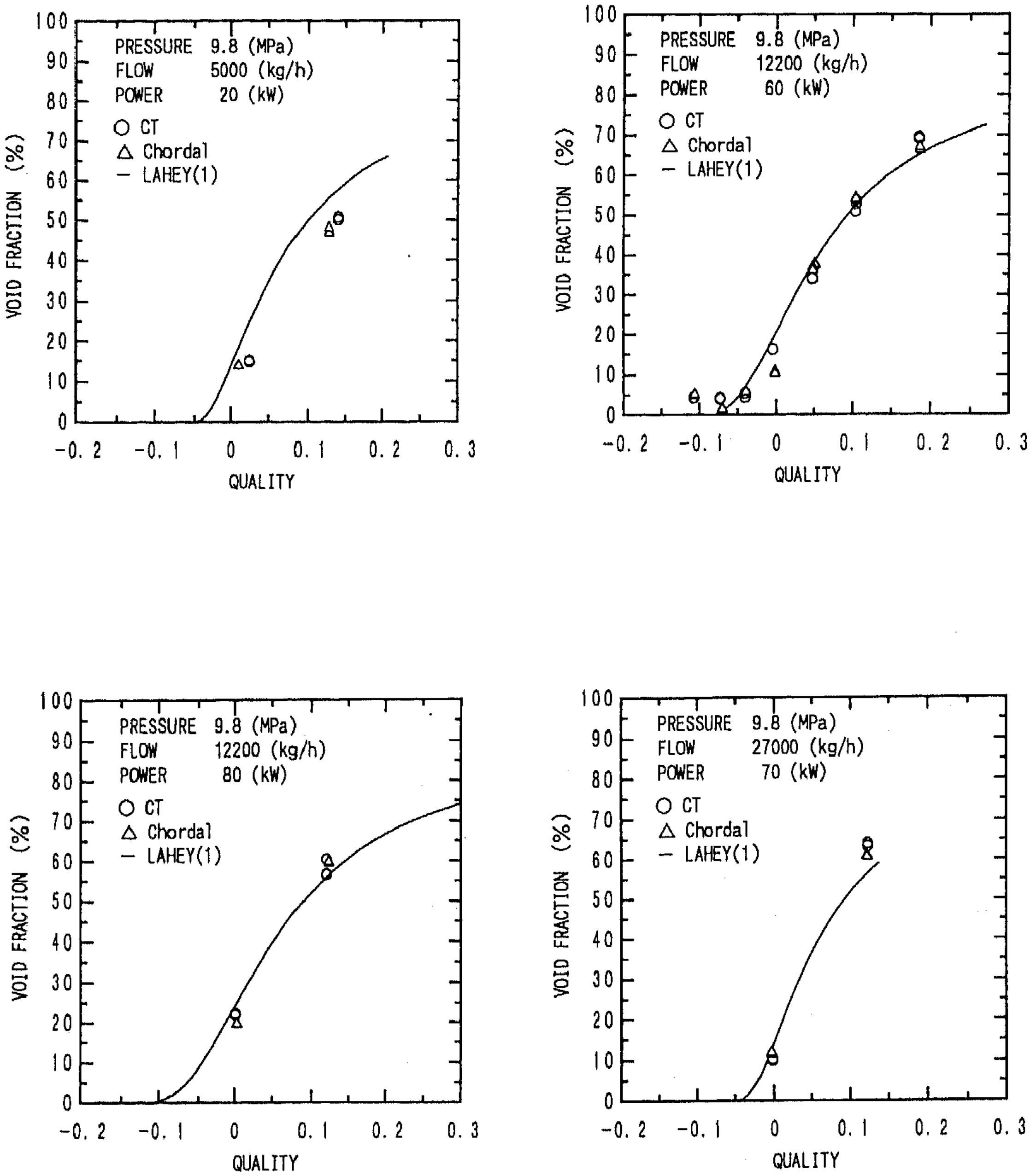



Figure 13 Measured Void Fraction vs Thermal Equibrium Quality
with Best Prediction (2) 9.8 MPa

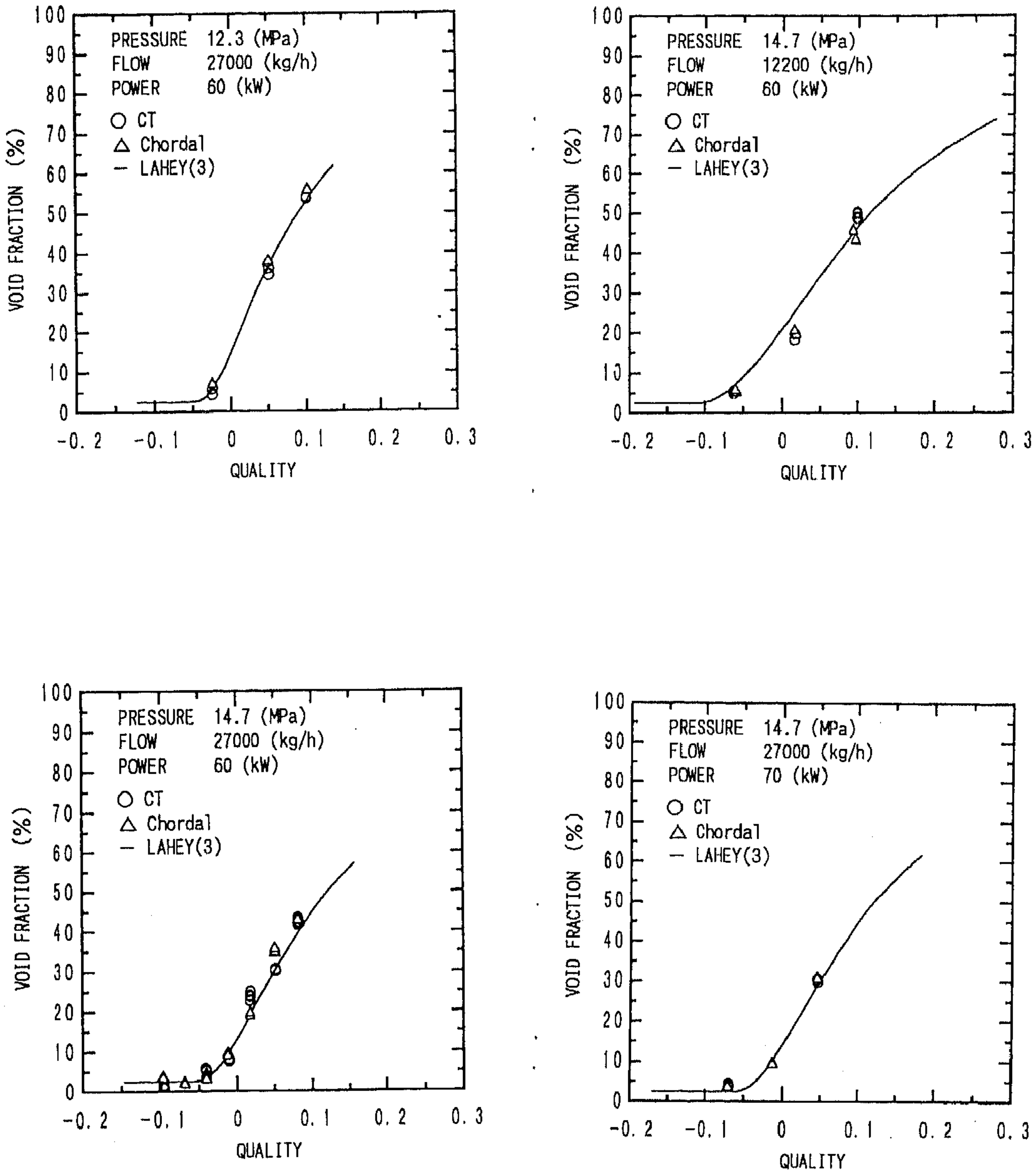



Figure 14 Measured Void Fraction vs Thermal Equibrium Quality
with Best Prediction (3) 12.3-14.7 MPa

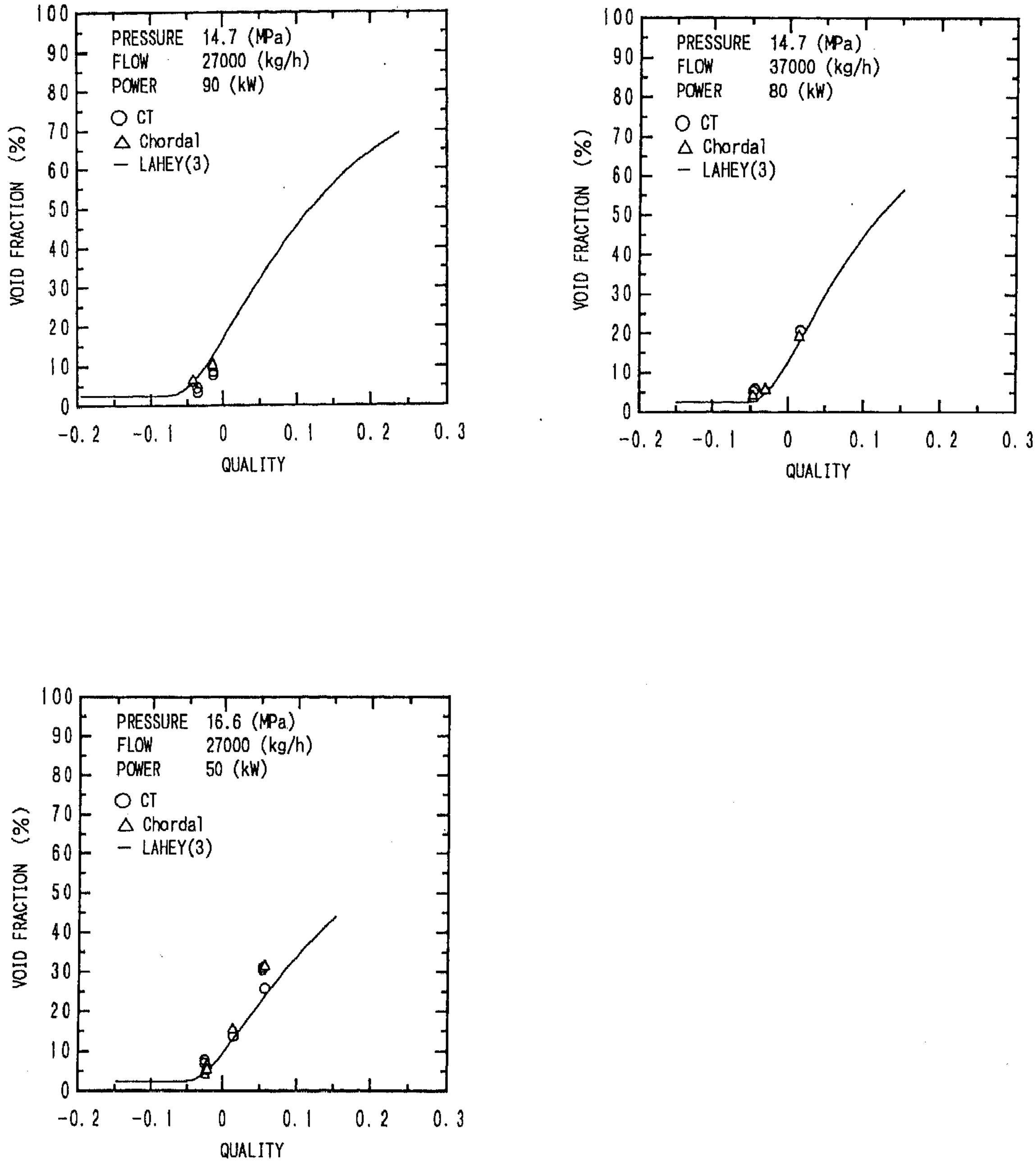



Figure 15 Measured Void Fraction vs Thermal Equibrium Quality with Best Prediction (4) 14.7-16.6 MPa

## NOMENCLATURE

L attenuation length, m
N count rate for attenuated beam, cps
$N_0$ count rate for incident beam, cps
$\alpha$ void fraction, -
$\mu$ mass attenuation coefficient, $m^2/kg$
$\rho$ mixture density, $kg/m^3$
$\rho_g$ density of gas, $kg/m^3$
$\rho_L$ density of liquid, $kg/m^3$

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